

1. Administrative & Study Identification

Full Study Title A Phase 3, Randomized, Open-Label Study of Nivolumab Combined With Ipilimumab Versus Sunitinib Monotherapy in Subjects With Previously Untreated, Advanced or Metastatic Renal Cell Carcinoma **Short Title/Acronym** CheckMate 214 **Protocol Number** CA209-214 **NCT Number** NCT02231749 **Sponsor Name** Bristol Myers Squibb **Investigational Product** Nivolumab combined with Ipilimumab **Phase of Development** Phase III **Trial Status** COMPLETED **Medical Monitor** Bristol Myers Squibb Clinical Operations / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the objective response rate (ORR), overall survival (OS), and progression-free survival (PFS) of nivolumab combined with ipilimumab versus sunitinib monotherapy in intermediate- and poor-risk participants with previously untreated advanced or metastatic renal cell carcinoma. **Secondary Objectives:** To evaluate investigator-assessed ORR, OS, and PFS in any risk participants, and to assess the overall safety profile and incidence of adverse events in all treated patients.

2.2 Background & Rationale To address the need for more effective, durable first-line treatments in advanced or metastatic clear-cell renal cell carcinoma. While targeted therapies like the VEGF-inhibitor sunitinib were standard of care, they rarely result in complete or durable responses. Dual immune checkpoint inhibition with nivolumab (anti-PD-1) and ipilimumab (anti-CTLA-4) has demonstrated complementary mechanisms of action, aiming to restore anti-tumor immune responses and improve long-term survival outcomes for patients with treatment-naïve advanced disease.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Sunitinib) **Blinding:** Open-label **Randomization:** 1:1, Stratified by International Metastatic RCC Database Consortium (IMDC) prognostic risk score (0 vs 1-2 vs 3-6) and geographic region.

3.2 Planned Enrollment & Duration Target **SN\$:** 1390 randomized patients **Number of Sites:** 175 hospitals and cancer centers across

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histological confirmation of renal cell carcinoma (RCC) with a clear-cell component. 2. Advanced (not amenable to curative surgery or radiation therapy) or metastatic (AJCC Stage IV) RCC. 3. No prior systemic therapy for RCC. Exception: One prior adjuvant or neoadjuvant therapy for completely resectable RCC is permitted if such therapy did not include an agent that targets vascular endothelial growth factor (VEGF) or VEGF receptors, and if recurrence occurred at least 6 months after the last dose. 4. Karnofsky Performance Status (KPS) of at least 70%. 5. Measurable disease as per Response Evaluation Criteria in Solid Tumors (RECIST) 1.1. 6. Tumor tissue [formalin-fixed paraffin-embedded (FFPE) archival or recent acquisition] must be received by the central vendor (block or unstained slides) in order to randomize a subject. (Note: Fine Needle Aspiration [FNA] and bone metastases samples are not acceptable).

4.2 Exclusion Criteria 1. Any history of or current central nervous system (CNS) metastases. Baseline imaging of the brain is required within 28 days prior to randomization. 2. Prior systemic treatment with VEGF or VEGF receptor targeted therapy (including, but not limited to, Sunitinib, Pazopanib, Axitinib, Tivozanib, and Bevacizumab). 3. Prior treatment with an anti-programmed death (PD)-1, anti-PD-L1, anti-PD-L2, anti-CD137, or anti-cytotoxic T-lymphocyte antigen 4 (CTLA-4) antibody, or any other antibody or drug specifically targeting T-cell co-stimulation or checkpoint pathways. 4. Any active or recent history of a known or suspected autoimmune disease or recent history of a syndrome that required systemic corticosteroids (>10 mg daily Prednisone equivalent) or immunosuppressive medications (except syndromes not expected to recur without external triggers). Subjects with vitiligo, type I diabetes mellitus, or residual hypothyroidism due to autoimmune thyroiditis only requiring hormone replacement are permitted. 5. Any condition requiring systemic treatment with corticosteroids (>10 mg daily Prednisone equivalents) or other immunosuppressive medications within 14 days prior to first dose of study drug. Inhaled steroids and adrenal replacement steroid doses >10 mg daily Prednisone equivalents are permitted in the absence of active autoimmune disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Arm A (Investigational):** Nivolumab 3 mg/kg IV + Ipilimumab 1 mg/kg IV every 3 weeks for 4 doses, followed by Nivolumab 3 mg/kg IV every 2 weeks. * **Arm B (Active Comparator):** Sunitinib 50 mg orally once daily for 4 weeks on, followed by 2 weeks off (6-week cycle).

Route of Administration: Intravenous (IV) for Nivolumab/Ipilimumab; Oral for Sunitinib. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or withdrawal of consent. (Treatment beyond progression permitted if investigator assessed clinical benefit was achieved).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care medications for symptom management. **Prohibited:** Concurrent systemic immunosuppressive therapy (unless utilized to treat immune-related adverse events), live vaccines, and other investigational anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, KPS, Baseline Tumor Imaging (including brain), Safety Labs
Baseline	Day 0	Randomization, First Dose Administration
Interim	Every 2-3 Weeks	Dosing, Safety Labs, Efficacy Assessment (Tumor scans Q12W), irAE monitoring
End of Study	Variable	End of Treatment Visit, Final Vital Signs, Exit Interview, Safety and Survival Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Investigator-assessed Objective Response Rate (ORR)** in Intermediate/Poor Risk Participants Per IRRC Using RECIST v1.1. 2. **Overall Survival (OS)** in Intermediate/Poor-Risk Participants With Previously Untreated Metastatic Renal Cell Carcinoma (mRCC). 3. **Progression-Free Survival (PFS)** in Intermediate/Poor-Risk Participants With Previously Untreated Metastatic Renal Cell Carcinoma (mRCC).

7.2 Secondary & Exploratory Outcomes 1. **Investigator-assessed Objective Response Rate (ORR)** in Any Risk Participants Per IRRC Using RECIST v1.1. 2. **Overall Survival (OS)** in Any Risk Participants With Previously Untreated Metastatic Renal Cell Carcinoma (mRCC). 3. **Progression-Free Survival (PFS)** in Any Risk Participants With Previously Untreated Metastatic Renal Cell Carcinoma (mRCC).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of clinical and laboratory AEs. Specific focus and algorithmic management protocols in place for immune-related adverse events (irAEs) associated with checkpoint inhibitors. **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Independent Data

Monitoring Committee instituted to assess safety signals and perform interim efficacy analyses.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Any Risk Participants for secondary efficacy endpoints; Intermediate/Poor-Risk set for primary efficacy endpoints; Safety Set for all patients who received at least one dose of study medication. **Sample Size Justification:** Approximately 1,390 patients powered ($\alpha = 0.05$, appropriate statistical power) to detect significant differences in the co-primary endpoints of ORR, OS, and PFS. **Handling of Missing Data:** Kaplan-Meier methodology used for time-to-event variables. Patients without documented progression or death were censored at the date of their last evaluable tumor assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence and data integrity. **Audit Trail:** Electronic Case Report Form (eCRF) implementation with centralized data management and blinded independent central review of imaging.

11. Study Identifiers & Governance

Primary ID: CA209-214 **Registry Number:** NCT02231749 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** Nivolumab Combined With Ipilimumab Versus Sunitinib in Previously Untreated Advanced or Metastatic Renal Cell Carcinoma (CheckMate 214) **Official Regulatory Title:** A Phase 3, Randomized, Open-Label Study of Nivolumab Combined With Ipilimumab Versus Sunitinib Monotherapy in Subjects With Previously Untreated, Advanced or Metastatic Renal Cell Carcinoma

12. Clinical Framework (Oncology/RCC Specific)

Disease Areas: Advanced Renal Cell Carcinoma, Metastatic Renal Cell Carcinoma **Diagnosis Threshold:** Histologically confirmed clear-cell RCC; IMDC Risk Group stratification **Medical Methodology:** First-line Systemic Therapy (Dual Immune Checkpoint Inhibition vs. Tyrosine Kinase Inhibitor) **Optimization Target:** Improvement of overall survival, progression-free survival, and durable objective response rate

13. Study Procedures & Methodology

Management Pathway: Standard oncological pathway for previously untreated mRCC **Education Protocol (HCP):** Extensive training on the rapid identification and algorithmic treatment of immune-related adverse events (irAEs) **Education Protocol (Patient):** Informed consent, recognizing and reporting immune-mediated symptoms early **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) / IRRC for standardized tumor efficacy readouts

14. Observation & Follow-Up Schedule

Total Duration: Minimum 42 months follow-up up to long-term survival follow-up (90+ months reported) **Onsite Visit Cadence:** Every 2 weeks or 3 weeks matching IV infusion schedules **Remote Monitoring Frequency:** Intermittent survival status tracking post-treatment discontinuation **Checklist Review Interval:** Routine tumor assessments performed every 12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				